

Scale or Shape? Decomposing NTK Movement During Grokking

Group G2

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1 Topic

Grokking is a phenomenon where the training loss of a neural network falls early but its test loss stays at chance for orders of magnitude longer before falling abruptly (Power et al., 2022). The grokking time t_{grok} counts the training steps from when the training loss first falls below a fixed level τ_{train} to when the test loss first falls below a fixed level τ_{test} . Grokking has been observed in networks trained on modular addition and, as such, it will be the task. The input concatenates the one-hot codes of $a, b \in \{0, \dots, p-1\}$ and the target is the one-hot code of $(a+b) \bmod p$. The neural tangent kernel (NTK) of a network $f(x; \theta)$ is $\Theta_t(x, x') = \langle \nabla_{\theta} f(x; \theta_t), \nabla_{\theta} f(x'; \theta_t) \rangle$, and K_t is its empirical version on a fixed probe set of n inputs. Training is *lazy* when Θ_t stays at Θ_0 , so the network behaves as its linearisation, and *rich* when Θ_t moves (Chizat, Oyallon, and Bach, 2019; Jacot, Gabriel, and Hongler, 2018). One account of grokking is a lazy-to-rich transition, where a linearised model memorises but generalising needs the kernel to move (Kumar et al., 2024; Mohamadi et al., 2024).

Saying that the kernel moves bundles two claims. A kernel can change in overall *scale*, with every eigenvalue rescaled and the eigenbasis fixed, or it can change in *shape*, so that the eigenbasis itself changes (e.g. rotation). Let $\hat{K} = K/\|K\|_F$ be the normalised kernel, and define the scale term $S_t = \log(\|K_t\|_F/\|K_0\|_F)$ and the shape term $R_t = 1 - \langle \hat{K}_t, \hat{K}_0 \rangle_F$. The standard measure of kernel change mixes the two, since it decomposes as

$$\|K_t - K_0\|_F^2 / \|K_0\|_F^2 = e^{2S_t} + 1 - 2e^{S_t}(1 - R_t). \quad (1)$$

The distinction matters because the theorem usually cited to link weight decay to kernel movement describes pure scaling. A network is k -homogeneous if scaling every parameter by c scales its output by c^k . A bias-free ReLU MLP has k equal to its depth. Lewkowycz and Gur-Ari (2020, Thms. 1–2) show that an infinitely wide k -homogeneous network trained with MSE and L_2 strength λ has kernel $\Theta_t = e^{-2(k-1)\lambda t} \Theta_0$. In our terms $S_t = -2(k-1)\lambda t$ and $R_t \equiv 0$, so the kernel shrinks but never changes shape and no features are learned. Yet Kumar et al. (2024, App. 13, arXiv v3) invoke this theorem to argue that weight decay ends lazy training, and hence that $t_{\text{grok}} \propto 1/\lambda$, an argument they call speculative and do not test on the kernel.

Research question. What sets t_{grok} ? The candidates are kernel shape, kernel or weight scale, and the weight-decay rate. Three 2026 accounts each back a different one (§2), and none measures shape change. We measure all three on one grid and ask which t_{grok} tracks.

Our hypothesis H is that shape change is the cause. Let Y be the probe set’s one-hot label matrix and $H = I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top$ the centring matrix. The cosine A_t between HK_tH and $HYY^\top H$ is centred alignment (Cortes, Mohri, and Rostamizadeh, 2012). Rescaling K_t leaves it unchanged, so only shape change moves it. H says generalisation begins when A_t crosses a threshold A_* , that A_* is shared across $\alpha, \eta\lambda, N$ and p , and that t_{grok} is independent of S_t given A_t .

The prediction that separates H from its rivals comes from the exact kernel evolution $\dot{\Theta}_t = -2(k-1)\lambda \Theta_t + T_t + T_t^\top$ (Lewkowycz and Gur-Ari, 2020, Eq. S5, arXiv v1). The weight-decay term only rescales Θ_t , so A_t moves only through T_t . As T_t vanishes at infinite width (Jacot, Gabriel, and Hongler, 2018; Lewkowycz and Gur-Ari, 2020), shape change is a finite-width effect and a wider network changes shape more slowly. Under H the target A_* is fixed, so t_{grok} is a distance over a rate and should grow with N and stay flat in $\eta\lambda$. We fit $\log t_{\text{grok}} = c + a \log N + b \log(\eta\lambda)$ on a crossed grid. Under H, $a > 0$ and $b \approx 0$. Under a weight-decay clock, $a \approx 0$ and $b \approx -1$. Two nonzero exponents mean two clocks, and the slower one sets the time.

2 Significance

Three 2026 accounts name three different clocks. Truong et al. (2026) intervene on the weight norm. Networks grok when the norm reaches a critical value $\|W\|_c \propto p^{0.38}$, and holding the norm at $\rho\|W\|_c$ makes t_{grok} grow exponentially in ρ , at a rate of about 7.5 for an MLP and 15 for unnormalised attention. Truong (2026) then shows that under cross-entropy this effect runs through the logit scale rather than the norm itself, but that this channel is absent under MSE, where about half of the effect is unexplained. Kim (2026) instead makes weight decay the clock, with $t_{\text{grok}} \propto (1 - \beta)/(\eta\lambda)$ where β controls heavy-ball momentum. All three are scale-like or optimiser-level quantities, so none can tell a reshaped kernel from a decaying one. H supplies the missing variable A_t . Our baseline uses MSE and vanilla gradient descent, where the logit-scale channel is absent, the theorem above applies as stated, and coupled L_2 regularisation coincides with decoupled weight decay.

The NTK-grokking link, the laziness knob α , eNTK spectral tracking, the norm delay law and norm mediation are all published. What is new here is decomposition (1) used as a timing variable, the (a, b) test that separates width-driven reshaping from the λ -clock, and a shape-based account of the norm law under MSE. If the timing of feature learning is set by kernel shape, then a quantity that is measurable during the apparently-flat phase of training predicts capability onset before any loss-based metric does, which provides an early-warning signal for delayed generalisation.

3 Feasibility

The baseline follows Kumar et al. (2024, App. 8.3, arXiv v3), so Tier 0 can be checked against a published figure. It uses one hidden layer of width $N = 100$, MSE loss, full-batch gradient descent at $\eta = 100$, $p = 23$, and 90% of the p^2 pairs for training. One setting of the swept hyperparameters is a cell, and the baseline is one cell. A cross-entropy arm at the baseline cell links our constants to Truong et al. (2026), whose results use cross-entropy.

Each checkpoint records the training and test MSE, the terms S_t , R_t and A_t on the centred kernel, the parameter movement, and $y^\top K^{-1}y$, which predicts when linearisation breaks. The empirical NTK on $n = 256$ probe points, summed over logits, takes seconds to run.

Three choices are fixed in advance. First, t_{grok} and its thresholds are pre-registered and sensitivity to τ is reported. A run has not grokked if its test loss is above τ_{test} when training stops. The run is kept, and t_{grok} is recorded only as longer than the run, which is right censoring. Second, laziness is set by the centred, rescaled predictor $\tilde{f} = \alpha[f(x, \theta) - f(x, \theta_0)]$ with $\eta = \eta_0/\alpha^2$ (Chizat, Oyallon, and Bach, 2019; Kumar et al., 2024). Since α , initialisation scale and label rescaling are one knob, we sweep only α . Third, weight decay is swept in $\eta\lambda$ below the stability bound $\eta\lambda < 2$.

The design has four arms. One is a 6×6 grid in $\alpha \times \eta\lambda$ at the baseline width. One is a 4×4 grid in $N \times \eta\lambda$, with N from 50 to 1600 and $\eta\lambda$ from 0 to 10^{-1} , for the (a, b) fit. One is a clamp arm with five values of ρ between 1 and 2, and one is a generality arm over three primes p and three training fractions. Five seeds per cell, and ten on the clamp arm where the MSE effect is half-size, give about 350 runs and roughly 20 GPU-hours.

The work falls into four tiers, each reportable on its own. Tier 0 reproduces the published α sweep and is a hard gate in week 1, since later work depends on it. Tier 1 measures S_t , R_t , A_t and t_{grok} over the grid, tests whether A_* is invariant, and exhibits a cell where kernel movement is large but A_t is flat. Tier 2, the head claim, is the (a, b) fit with bootstrap intervals and the residuals where a power law fails. Tier 3, a stretch goal, replicates the clamped-norm law under MSE and tests whether A_* is invariant to ρ .

References

- Chizat, L., E. Oyallon, and F. Bach (2019). “On lazy training in differentiable programming”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. arXiv: 1812.07956.
- Cortes, C., M. Mohri, and A. Rostamizadeh (2012). “Algorithms for learning kernels based on centered alignment”. In: *Journal of Machine Learning Research* 13.28, pp. 795–828.
- Jacot, A., F. Gabriel, and C. Hongler (2018). “Neural tangent kernel: Convergence and generalization in neural networks”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. arXiv: 1806.07572.
- Kim, T. (2026). “Grokking on the weight-decay clock: A rate hierarchy from softly broken symmetries”. In: *arXiv preprint*. arXiv: 2607.23967.
- Kumar, T. et al. (2024). “Grokking as the transition from lazy to rich training dynamics”. In: *International Conference on Learning Representations (ICLR)*. arXiv: 2310.06110.
- Lewkowycz, A. and G. Gur-Ari (2020). “On the training dynamics of deep networks with L_2 regularization”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. arXiv: 2006.08643.
- Mohamadi, M. A. et al. (2024). “Why do you grok? A theoretical analysis of grokking modular addition”. In: *International Conference on Machine Learning (ICML)*, PMLR 235. arXiv: 2407.12332.
- Power, A. et al. (2022). “Grokking: Generalization beyond overfitting on small algorithmic datasets”. In: *ICLR MATH-AI Workshop*. arXiv: 2201.02177.
- Truong, X. K. (2026). “What does the weight norm control in grokking? Logit-scale mediation under cross-entropy”. In: *arXiv preprint*. arXiv: 2606.18465.
- Truong, X. K. et al. (2026). “The weight norm sets the grokking timescale: A causal delay law”. In: *arXiv preprint*. arXiv: 2606.13753.